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Supply function competition with CEO bias

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ABSTRACT

We investigate the motivation for a firm to hire a biased chief executive officer (CEO) under supply function competition. Previous work has found that in symmetric duopoly models such as Cournot and Bertrand, it is optimal for owners to hire overconfident CEOs. Recent literature has revealed, however, that many firms compete in supply functions. We develop a two-stage model in which firm owners first decide to hire CEOs who are overconfident, underconfident, or rationally confident in their estimation of supply or demand. The potentially biased CEOs then compete in supply functions. We find that, with demand bias, it is optimal to hire CEOs who underestimate demand. With the potential for supply bias, however, it is optimal to hire CEOs who are unbiased. Thus, an owner's decision depends on both the source of potential bias and the mode of competition.

KEYWORDS

Behavioural economics; firm objectives; CEO bias; supply function competition

JEL CLASSIFICATION

D21; D43; L13; L21

I. Introduction

Many CEOs of large corporations are overconfident.¹ A theoretical literature establishes conditions under which CEO overconfidence may arise. In non-strategic settings, such as monopoly or perfect competition, it is optimal for owners to hire unbiased CEOs. However, in oligopolistic markets where strategy is important, several studies find that CEO overconfidence can be optimal (e.g. Campbell et al. 2011; Englmaier and Reisinger 2014). Underconfident (or more conservative) CEOs are also observed (Bertrand and Schoar 2003; Huang and Kisgen 2013). Schroeder, Tremblay, and Tremblay (2022) extend the theoretical literature to allow for underconfidence.

We examine a type of competition that has yet to be considered in the overconfidence literature: competition in supply functions. Instead of choosing a single price or quantity, a firm chooses a function that specifies the quantity it will produce at any price. Understanding CEO confidence bias in this context is crucial. A growing literature finds that supply function competition is both common and different in important ways from the standard oligopolies under which CEO bias has been studied. New theoretical results are required to gain insight into firm strategy in this setting.

We find that whether firm owners can profit by hiring biased CEOs depends on the source of potential bias. These results contrast with the previous literature and demonstrate that the motive for hiring a biased CEO can be complex in strategic settings.

II. Literature

Supply function as a strategic choice was first introduced by Grossman (1981) and developed by Klemperer and Meyer (1989) and Vives (2011). The empirical literature has found many markets in which firms compete in supply functions. The leading application, pioneered by Green and Newbery (1992) for the case of Great Britain, is wholesale electricity markets. Generator companies compete in reverse auctions to sell to a power company, submitting supply schedules that disclose the quantity supplied for all feasible market prices. Recently, supply function models have been used to analyse wholesale energy markets in the US (Qiu 2020), India (Ryan 2021), and Australia (Neill 2022).

An active theoretical literature explores the properties of supply function competition models and how they compare to those of Cournot and

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¹For a review, see Malmendier and Tate (2015).

Bertrand competition. Existing work compares equilibrium profits and welfare across models (Delbono and Lambertini 2016; Saglam 2018) and explores the differential impacts of regulation (Saglam 2025). This literature has not yet incorporated CEO confidence bias.

Schroeder, Tremblay, and Tremblay (2022) demonstrate that firm owners benefit from hiring overconfident CEOs in duopoly settings where both firms' choice variables are strategic substitutes (as in Cournot) or strategic complements (as in Bertrand). When the choice variables are strategic substitutes for one firm and strategic complements for the other firm (as in Cournot-Bertrand), it is optimal for owners to hire underconfident CEOs.

Although the choice variables in supply function competition are strategic complements, we find that the previous result from Bertrand competition – that overconfident CEOs are always optimal – does not apply. As detailed in Section II, an owner's decision to hire a biased CEO can profitably serve as a commitment device when the potential for bias is introduced through misestimation of market demand. When the bias is introduced into supply, it is not profitable to hire a biased CEO.

III. Model

Consider an industry with two firms (1 and 2) that produce homogeneous goods and compete in a two-stage game of perfect and complete information. In stage I, owners maximize profit (π) by deciding whether to hire a biased CEO. In the second stage, each CEO chooses a supply function to maximize expected profit based on their degree of bias. A firm's supply function defines a contract with buyers that legally obligates the firm to supply a particular quantity at any given price.

The structure of the model follows Ciarreta and Gutierrez-Hita (2006). Absent bias, market demand is $Q_D(p) = a - bp$, where Q_D is quantity demanded, p is price, $a \in (0, \infty)$, and $b \in (0, \infty)$. Firm i 's supply function is $q_i(p) = s_i p$, where q_i is quantity supplied. The supply parameter $s_i \in (0, \infty)$ is the choice variable of the firm.² Industry supply is

$Q_S(p) = s_1 p + s_2 p$. Firm i 's total production cost is $(c/2)q_i^2$, where c is a positive constant and $(a - c)$ is positive and sufficiently large to assure firm participation. There is an added fixed cost of identifying and hiring a biased CEO: $f(\phi_i) = (1 - \phi_i)^2$, where ϕ is a non-negative index of CEO bias. This specification implies that it is increasingly costly to hire a CEO with a greater degree of bias. When the CEO of firm i (CEO $_i$) is unbiased, $\phi_i = 1$; when overconfident, $\phi_i \in (1, \infty)$; when underconfident, $\phi_i \in (0, 1)$. Total cost is $TC_i = (c/2)q_i^2 + (1 - \phi_i)^2$.³

The potential for bias occurs in one of two ways. In the first, overconfident (underconfident) CEOs overestimate (underestimate) market demand. Second, CEOs may choose a biased supply-function contract, where an overconfident (underconfident) CEO overestimates (underestimates) the firm's ability to supply output at a given price.

Demand bias

Demand bias is the most common form of bias in the literature. Here, CEO $_i$ maximizes expected profit based on their potentially biased forecast of demand: $E_i(Q_D) = \phi_i a - bp$. CEO $_i$ is unbiased when $\phi_i = 1$, but overestimates (underestimates) demand when $\phi_i > 1$ ($\phi_i < 1$).

We use backwards induction to determine the subgame-perfect Nash Equilibrium (SPNE), with its formal derivation found in the Appendix. The SPNE value of the variable of interest, the optimal level of bias, is:

$$\phi_i^{SPNE} = \frac{(4 + bc)[4by_i + a^2(y_i - y_j)]}{4b(4 + bc)^{3/2} - a^2(3 + bc)(4y_j + b^{3/2}c^{3/2} - (2 + bc)y_i)}, \quad (9)$$

where subscript j identifies firm i 's competitor. Given the range of parameter values defined above, $0 < \phi_i^{SPNE} < 1$: when firms compete in supply functions, it is optimal for owners to hire CEOs who underestimate demand.

The intuition stems from the fact that the underestimation of demand causes CEO $_i$ to supply less output (behave less aggressively). When the owner of firm i considers hiring a biased CEO in stage I,

²With a non-zero intercept, the equilibrium is invariant to optimizing with respect to the intercept or the slope parameter (Delbono and Lambertini 2015). If $s_i \neq 0$, the equilibrium values of the supply intercepts will equal 0 (Yasui and Haraguchi 2018).

³The second-order conditions are met for all choice variables.

$\partial \pi_i^{\text{owner}} / \partial \phi_i < 0$ and $\partial \pi_j^{\text{owner}} / \partial \phi_i < 0$ when these derivatives are evaluated at $\phi_i = \phi_j = 1$. If CEOs are unbiased, both owners benefit when firm i hires a CEO who underestimates demand. From an owner's perspective, hiring such a CEO is commitment device that raises firm profits.

This result is the opposite of that found in symmetric Cournot and Bertrand models, where firms face a Prisoners' Dilemma and it is always optimal for owners to hire CEOs who overestimate demand. Here, producing less output benefits both firms. The difference in results is consistent with evidence that more conservative CEOs are found in public utility industries.

Supply bias

In this specification, demand is estimated accurately, but supply may be estimated with bias. CEO _{i} estimates the supply function to be $q_i = \bar{s}_i p$, where $\bar{s}_i = \phi_i s_i$. CEO _{i} is unbiased when $\phi_i = 1$. Overconfidence occurs when $\phi_i \in (1, \infty)$, which means that CEO _{i} believes that more output can be profitably supplied at a given price. With underconfidence, $\phi_i \in (0, 1)$. The resulting expected market supply is $E_i(Q_S) = \bar{s}_1 p + \bar{s}_2 p$.

We use backwards induction to identify the SPNE, which is formally derived in the Appendix. Given the anticipated stage II outcome, the owner's profit falls as ϕ_i deviates from 1. Thus, $\phi_i^{\text{SPNE}} = 1$; it is optimal for owners to hire unbiased CEOs.

Owners cannot benefit from hiring biased CEOs in this case because bias has no effect on the CEO's optimal choice of $\bar{s}_i$ in stage II. A marginal increase in ϕ_i leads to a proportional decrease in s_i so that $\bar{s}_i^*$ remains the same. Thus, bias has no effect on the true equilibrium values of output or price. Given that deviating from $\phi_i = 1$ raises costs by $(1 - \phi_i)^2$ and has no effect on a firm's revenue or production cost, it is optimal for owners to hire unbiased CEOs.

Demand versus supply bias

If the owner of firm i can choose the type of CEO bias, the discussion demonstrates that it is optimal

to favour demand bias. Hiring a CEO who underestimates demand raises owner profit, while any supply bias lowers profit.

IV. Discussion

Previous studies find that it is optimal for owners to hire overconfident CEOs when choice variables are either strategic complements or strategic substitutes. Our analysis provides two exceptions, showing that it is optimal to hire either underconfident or unbiased CEOs, even though choice variables are strategic complements, with supply-function competition.

The fact that an owner's decision depends on the source of potential bias and the mode of competition suggests that we should see variation in CEO bias across industries of different types. Chen et al. (2019) find that CEO confidence is industry-specific, with more conservative CEOs employed in the public utility and telecommunication industries. The literature does not include measures of CEO confidence in the energy sector specifically, where supply-function competition is common, but future work in this area could provide empirical support for our findings.

V. Conclusion

We study a duopoly model where owners can hire biased CEOs and firms compete in supply functions. We find that whether it is optimal to hire biased CEOs depends on the source of bias. With demand bias, it is optimal to hire CEOs who underestimate demand. Owners use this strategy as a commitment device to extract higher profits. With the potential for biased estimation of supply, however, it is optimal for owners to hire unbiased CEOs.

The contrast between these and previous results on CEO bias suggests avenues for future work, for example, examining the implications of CEO bias in supply function competition on long-run outcomes such as market stability and consumer welfare.

Author contributions

CRedit: **Hoang Pham:** Conceptualization, Formal analysis, Investigation, Writing – original draft; **Elizabeth Schroeder:** Conceptualization, Formal analysis, Investigation, Writing –

original draft; **Victor J. Tremblay**: Conceptualization, Formal analysis, Investigation, Writing – original draft.

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